CH 4 - Competition

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Review - Gain from trade:

- Consumers with endownments can become happier by trading $\rightarrow$ Expands total market utility
- An observation was the tendency for consumers with lower endownments to gain more from trading
- Relatives prices emerge

Example:

Consider a world with two countries A and B. Both countries produce item x and item y.

A can produce at most 60 item x or 120 item y, or any linear combinaiton of the two

B can produce at most 30 item x or 90 item y, or any linear combination of the two.

Both countries spend half their time producing item x and the other on item y

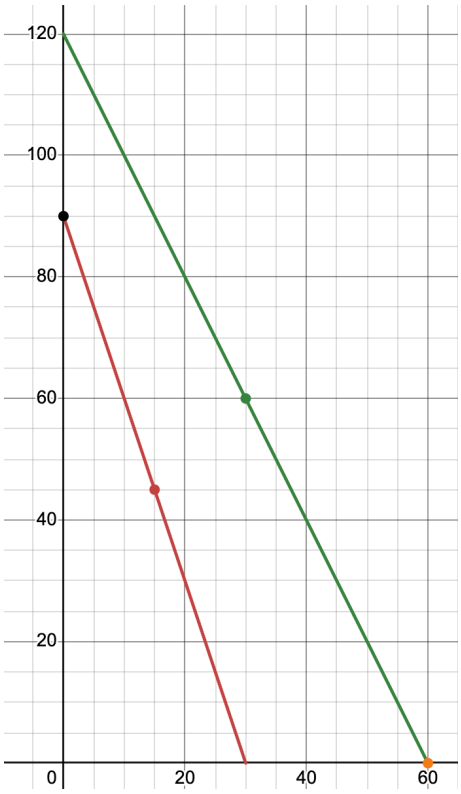
Country A produces 30 item x and 60 item y

Country B produces 15 item x and 45 item y

Total: 45 item x, 105 item y

🐳 **Info — Definition**

Comparative Advantage : Having a lower opportunity cost than everyone else.



Country A costs $\frac{1}{2}$ item x to produce item y; Country B costs $\frac{1}{3}$ item x to produce item y

Let A only produce item x and B only produce item y

Total: 60 item x, 90 item y then they trade

A will sell x for 2 y or higher

B will sell 3 y for 1 y

Price range $[2, 3]$ y per x

Pick a feasible price ratio to demonstrate: $2.5 : 1$ y per x

A

	0	1	2	3	...	24
x	60	59	58	57	...	36
y	0	2.5	5	7.5	...	60

B

	0	1	2	3	...	24
x	0	1	2	3	...	24
y	90	87.5	85	82.5	...	30

Notice that the total production of x increases and y decreases due to **Absolute Advantage** that A has.

Let A produce 15 more y, 7.5 of x is lost.

Total production : 52.5 x and 105 y on market

A

	0	1	2	3	...	18
x	52.5	51.5	50.5	49.5	...	34.5
y	15	17.5	20	22.5	...	60

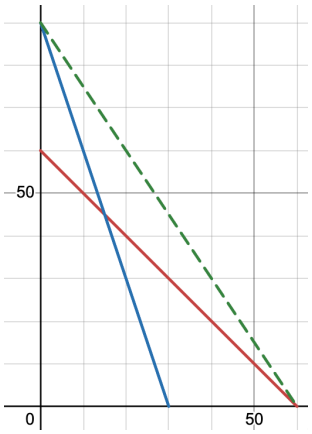
B

	0	1	2	3	...	18
x	0	1	2	3	...	18
y	90	87.5	85	82.5	...	45

All countries get an increase in x

Gains from trade fro specialization, we will get the country with absolute advantage to produce just enough y that the post-specialization production equation pre-specialization.

Then trade.



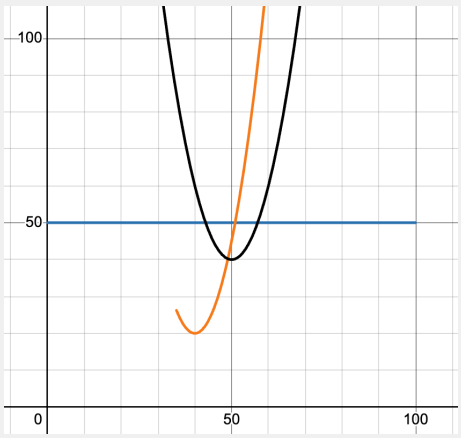
Comparative advantage is all needed for gains from trade

- So long as PPFs(green dotted line) are not parallel, there is gains from trade.
- Absolute advantage limits how much we can specialize

🐳 **Info — Competitive Model**

- So called be cause we use it to study different market structures: Competitive $\leftrightarrow$ Non-competitive
- Fundamentally, all of our model have been following the same idea:
  - Marginal Benefit $>$ Marginal Cost
- Firms & Companies exist to make profit:
  - Revenue $>$ Cost
- Marginal Revenue $\sim$ Price P
  - Price depends on market structures (i.e. Number of firms, Easy with which firms can enter the market, Easy with which products can be differentiable, Market information, Friction, etc.)
- Marginal Cost(MC)
  - Cost function $C(q) = Aq^2 + Bq + C$
  - Fixed cost: does not scale with production
  - Variable cost: scale with production
  - $C(q)$ change if q goes up by 1.
- Optimal point: $P = MC$
- Average cost: $\frac{C(q)}{q}$

Graph (Assume Perfect Competition)



Blue: price
Orange: marginal cost
Black: average cost

Example:

$C(q) = 700 + 12q + 5q^2$

Marginal Cost: $C'(q) = 12 + 10q$

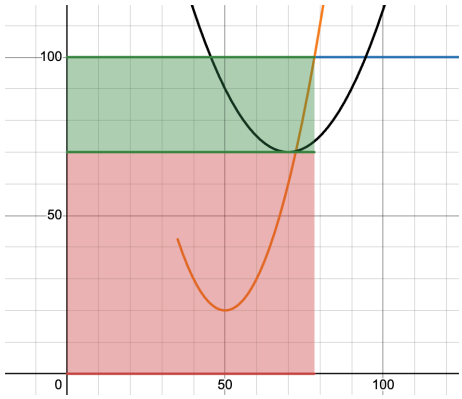
Let $p = 250$, $MR = MC \implies p = MC \implies 250 = 12 + 10q \implies q = 23.8$

1. Long-run Competitive ($P > AC$)
   $\pi(\text{Profit}) = \text{Revenue} - \text{Cost}$

   $= pq - (700 + 12q + 5q^2) \implies 250(23.8) - (700 + 12(23.8) + 5(23.8)^2) = 2132.2$

   $AC(q^*) = \frac{C(q^*)}{q^*} = 160.41$

   $(250 - 160.41) * 23.8 = 2132.2$



   Orange: profit
   Green: Cost

   MC intersects AC at AC's lowest point

   Note that the minimum of average cost is meaningless

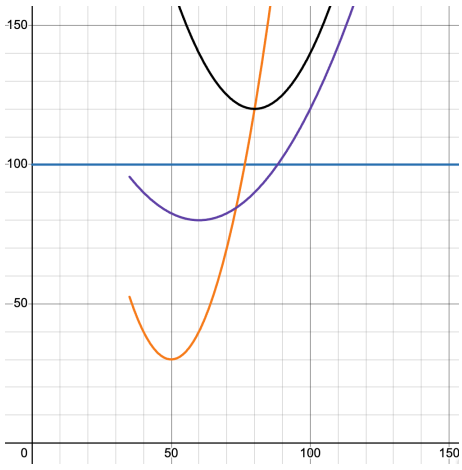
2. Short-run Competitive ($AC > P > AVC$) Observe that the cost of producing nothing is higher than the cost of producing something.

   So $C(0) = 700, \pi(0) = 0 - 700 = -700$

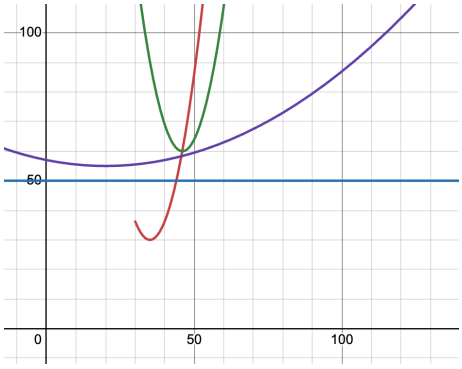
   But then, producing at the optimal is $\pi(q*) = -300 > -700$, losing less if producing at the optimal then producing nothing.

   Theefore, some existing companies, producing while lossing money can benefit the companies from lossing less.

   So, Average Variable Cost: $\frac{VC}{q} = 12 + 5q$



3. Shutdown ($AC > AVC > P$)



Long-run means that we can change anything

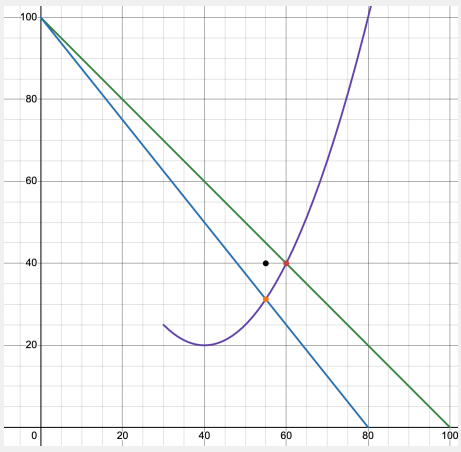
Short-run means we can only change variable cost

The firm always produces at $p = MC$

🐳 **Info — Monopolistic Production**

- One producer/one opponent producer
- Non-competitive market
- They can set price with a limit of the demand side of the market.
  - Marginal Benefit > Marginal Cost
  - Marginal Revenue > Marginal Cost
  - Marginal Revenue $= p - \Delta p(q - 1)$, is also the downward rotation of demand

Model:



Green: Demand
Blue: Marginal Revenue
Purple: Marginal Cost
Optimum at black dot

Monopoly: use Demand = MR

Competitive: use p = MR